



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.12

TUTOR GUIDE

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RELEVANT PRIOR LEARNING

MED 1000

- 1.30 Syndromes of alcohol misuse [L]
- 1.30 Pharmacokinetics and metabolism of alcohol [L]
- 1.32 Benzodiazepines [L]

MED2000

- 2.16 Assessment & management of the psychotic patient
- 2.17 Assessment and management of overdose
- 2.25 Mental Health Workshop (CCS)

MED 3000

- 3.13 Alcohol misuse and withdrawal. [Short case tutorial]
- 3.13 Substance abuse disorders [Short case tutorial]

WEEK 12_ EMERGENCY MEDICINE		
YR	Wk	LEARNING OBJECTIVES
4	12	• Demonstrate an understanding of the causes of acute agitation by be able to assess a patient with this clinical presentation. • Outline an investigation plan in the assessment of the acutely agitated patient and be able to interpret the results in the context of the clinical presentation. • Outline the principles and key priorities of management of the acutely agitated patient.
4	12	• Know the key components of the history and examination of the acutely intoxicated patient. • Outline an investigation plan in the assessment of the acutely intoxicated patient and be able to interpret the results in the context of the clinical presentation. • Know the key components of the assessment of the acutely intoxicated patient with a deteriorating level of consciousness and be able to outline the possible causes for this presentation.
4	12	• Outline the typical clinical presentations of overdose due to heroin, cocaine, amphetamines, Gamma Hydroxy Butrate [GHB], ecstasy and benzodiazepines. • Outline the typical clinical presentations overdose due to tricyclics, antidepressants, monoamine oxidase inhibitors [MAOI] and Selective serotonin reuptake inhibitors [SSRI] • Outline the role and mechanism of action of activated charcoal in the management of overdose. • Outline the role of dialysis in the management of suspected overdose or poisoning.
4	12	• Demonstrate an understanding of the possible causes of acute hyperthermia by being able to outline the key components of the assessment of the patient presenting with hyperthermia. • Outline the immediate management of the patient presenting with acute hyperthermia. • Summarise the investigations in the assessment of the patient with acute hyperthermia and be able to interpret the results in the context of the clinical presentation.

CASE ONE

Short case number: 4_12_01

Category: Clinical Skills/ Mental health & human behaviour

Discipline: Emergency Medicine

Setting: Emergency Department- Tertiary Hospital

Topic: The Acutely Agitated patient

CASE



You are a resident working in an Emergency Department. The police bring in a handcuffed agitated patient. They give the story that he was found wondering the streets being aggressive towards everyone he met. The police state that he has been aggressive and violent and that it has taken several officers to control him.

In your department, he is handcuffed on a trolley, with 2 police and 3 security staff restraining him. He is yelling and spitting, not answering your questions and not making sense with what he is saying.

QUESTIONS

1. Outline possible causes for this patient's agitation
2. Outline important aspects of the history that you would pursue to assist you in making a diagnosis.
3. What investigations could you perform, to assist you in determining the diagnosis?
4. Outline the various methods for managing the agitated patient.
5. What are your priorities in this case?

Resources:

- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

1. Outline possible causes for this patient's agitation

There are many potential causes of this patient's agitation, the most important ones are:

- drug related causes ie. illegal substances that may be causing this behaviour directly or resulting in an acute psychosis. Drug related causes would include; alcohol, amphetamines, ecstasy, LCD, cocaine etc.
- Is there agitation secondary to a trauma related head injury? Patients with head injury may be in an agitated state and their behaviour may be misconstrued for something else
- Is this an acute or an exacerbation of a chronic psychiatric illness?
- There may be rarer causes that can cause some confusion and potential aggression, for example an acute decreased sodium level, or acute hypoxia, or an arrhythmia causing hypoxia.
- Infective processes such as an encephalitis, or neurological sequelae of endocarditis

2. Outline important aspects of the history that you would pursue to assist you in making a diagnosis.

Any history that can be collected will prove to be very helpful.

- Has the patient been identified? The patient can then be looked up on the hospital computer, or the psychiatric team can look at their database to ascertain if the patient has a past history.
- It is important to ascertain if there has been any past history of this behaviour? This may give a clue as to history of illicit drug use, or psychiatric conditions
- Does the patient have a known psychiatric history?
- Was there anyone with the patient? Family or friends can provide valuable information in terms of the presenting complaint as well as the past history. They can discuss aspects of any drug-taking behaviour etc.
- Any past medical history is important.
- The exact behaviour of the patient before he was picked up is important. Was the patient only aggressive, was the patient making sense? Has his condition changed in any way? ie. has the patient become more incoherent, or making less sense now. Has the conscious state changed in any way?
- Has the patient received any head trauma, either prior to the police apprehending, or as a consequence of their apprehension. Beware to ensure that the patient's condition hasn't changed secondary to accidental head strike during apprehension. If the patient's conscious state is changing, we need to be determining if this is secondary to the any drugs including alcohol, or if it is due to intracranial pathology.
- Will need to look at in the patients pockets (carefully) as there may be evidence of drugs etc..

3. What investigations could you perform, to assist you in determining the diagnosis?

The investigations to consider would be immediate investigations, or bedside tests and then other investigations.

- Bedside tests would include a blood sugar level and an ECG. Need to ensure that the patient has a normal range BSL and a lower BSL may cause altered behaviour. Also need to ensure that the ECG is not abnormal and that the patient is not affected secondary to an arrhythmia.
- Pathology –
- Full Blood Count, looking for infective causes, or a low platelet count that may point to intracranial haemorrhage
- Electrolytes – These include sodium abnormalities, or renal impairment
- Liver Function tests
- Coags – they may give some evidence as to a coagulopathy and intracranial bleed.
- Blood alcohol levels will give an idea of the level of alcohol intoxication
- Paracetamol and salicylate levels – these are important if the patient has a psychiatric history and there is any concern of an overdose.
- Urine drug screen – although not acutely helpful may be of assistance in the long term management of this patient.
- Consider Head CT in this patient if there is any concern of intracranial pathology. Remember in trauma, to also manage the cervical spine.

4. Outline the various methods for managing the agitated patient.

The immediate management of this patient is about keeping both the patient and the staff safe. In an agitated patient it is important to ensure that the patient does not injure themselves but also that they do not injure the staff.

There must also be a state achieved where investigations can be carried out on the patient, again so that they are accurate and secondly so that they are performed safely. Examples of this include taking blood tests whilst minimising the risk of needle stick injury and secondly.

Immediate management of the patient includes calming the patient. If it is believed that this is a drug or alcohol dependant aggression, then rapid control of the patient with simple sedation may be obtained. A sedative agent such as midazolam can be used. In the patient where it is difficult to obtain IV access, the sedation can be given intramuscularly and when the patient is significantly calmer then access can be obtained and bloods sent.

If patient not restrained- consider 5 point 'take-down'.

An urgent blood sugar should follow, with a view to correcting this if necessary.

There should then be the equivalent of a primary survey done on the patient. If head injury is suspected, there may be a concern with sedating the patient as it may be difficult to then monitor the conscious state, however definitive investigation takes precedence and if the conscious state deteriorates such that the GCS is <9 then intubation would need to be considered. Also if head injury is suspected then the cervical spine must be immobilised and imaged, appropriately with a CT scan at the same time as the head CT.

If the examination is normal, an ECG should be obtained.

5. What are your priorities in this case?

Priorities are safety of the patient and the staff and making a rapid diagnosis to ensure the patient does not have an acutely reversible, dangerous condition.

CASE TWO

Short case number: 4_12_02

Category: Clinical Skills / Mental Health & Human behaviour

Discipline: Emergency Medicine

Setting: Emergency Department, St Vincents Hospital

Topic: Acute Intoxication

CASE



You are on night shift in the Emergency Department at St Vincent's Hospital. Ambulance officers bring in a 21 year old male, Jasper, with a decreased level of consciousness. His friends called an ambulance as he was drowsy, vomiting, and unable to stand.

They state he has had "heaps" to drink tonight, and are evasive when asked about recreational drug use. They don't know of any medical conditions or medications that the patient is on. He drinks heavily both nights of most weekends. There was a period of time where they were separated from him, and they state he "seemed ok" prior to that.

Jasper is lying in the recovery position on the bed. He smells strongly of alcohol and vomitus. His observations are:

HR 110

BP 106/55

RR 12, SaO₂ 96% RA

GCS 9/15 (E2V3M4)

QUESTIONS

1. Outline your immediate assessment and management of this patient.
2. What investigations are appropriate in this patient?
3. Can you be sure that his decreased level of consciousness is due to alcohol intoxication? Discuss the possibilities for decreased LOC in this patient, and how you would differentiate between these.

FURTHER CASE INFORMATION

You receive a call from the pathology lab. Jaspers blood alcohol level is 55mmol/L.

QUESTIONS

4. Outline the various methods of measuring blood alcohol level/content. Does this confirm that Jasper's decreased LOC is due to alcohol?
5. Following your assessment and investigations, you're diagnosis is alcohol intoxication. What is an appropriate plan of management for this patient? Include discussion of appropriate disposition of this patient.
6. Are you concerned about the amount that Jaspers drinks? What types of intervention are available for this patient population? How would you approach this issue with Jasper?

ANSWERS

1. Outline your immediate assessment and management of this patient.

ABCDE Primary Survey / Assessment / management

- Particularly important to assess airway, and level of consciousness as severely intoxicated individuals are at risk of aspiration/loss of airway due to decreased level of consciousness +/- vomiting. Patients may need a definitive airway provided, i.e. intubation.

2. What investigations are appropriate in this patient?

- Blood tests include EUC, FBC, LFT, CMP, blood alcohol level at minimum
There was a period when the patient was unobserved, and it is possible that he has fallen or sustained some form of head trauma. Careful examination +/- extra history can help determine whether CT Brain is required in this patient to ensure that his decreased level of consciousness is not a result of acute intracranial pathology.

3. Can you be sure that his decreased level of consciousness is due to alcohol intoxication? Discuss the possibilities for decreased LOC in this patient, and how you would differentiate between these.

No. We can never assume that Decreased LOC is purely a result of alcohol or other drug intoxication. See below table for other conditions to consider in presentation of Intoxication + Decreased LOC.

Table 21.4.2: Illnesses not to be missed in the presumed intoxicated person

Metabolic and Encephalopathic	Hypoglycaemia Hyperglycaemia Wernickes Hyponatraemia Liver failure Renal Failure
Head Injury	Skull fracture Cerebral Contusion Subdural or extradural haemorrhage / haematoma
Other intracranial pathology	Infection CVA Seizure / Post-Ictal State Space-occupying lesion
Toxicological - CNS depressants - Illicit Drugs - Prescription Medications	Opiods Ecstasy, GHB, Ketamine Benzodiazepines, Anticonvulsants

- Other alcohols	Methanol, ethylene glycol, isopropyl alcohol
Sepsis	CNS, UTI, Pneumonia, aspiration

4. Outline the various methods of measuring blood alcohol level/content. Does this confirm that Jasper's decreased LOC is due to alcohol?

- Portable Breathalyser – estimates Blood alcohol concentration (BAC) by measuring alcohol concentration of alveolar air. Approximate difference of 15-20% between breath alcohol readings and serum blood alcohol concentration. Readings influenced by: temperature, haematocrit level, other substances (i.e. Ketamine), hyper or hypo-ventilation prior to exhalation. There is also the possibility of machine error.
- Blood Serum Levels – measures qualitatively the concentration of alcohol in blood

ETOH Dose (g/kg)	SI Units	Mg/dL	g/DL (%)	Clinical Features
0.5	11	50	0.05	Disinhibition and Euphoria
1	22	100	0.10	Slurred Speech, significant CNS depression in non-tolerant individuals
2	43	200	0.20	Potential for coma, ETOH dependent individual patients usually ambulant
>5	>87	>400	>0.4	Coma, respiratory depression, hypotension

5. Following your assessment and investigations, you're diagnosis is alcohol intoxication. What is an appropriate plan of management for this patient? Include discussion of appropriate disposition of this patient.

- Supportive Care - ABCDE
- Monitoring – Vital signs, GCS, Neurological observations
- Opportunistic Alcohol Screening + Brief Intervention

6. Are you concerned about the amount that Jaspers drinks? What types of intervention are available for this patient population? How would you approach this issue with Jasper?

- Conduct an Alcohol misuse screening test: CAGE, Paddington Alcohol test (PAT).
- Brief Intervention / Counselling: 10-15min intervention providing information on recommended safe drinking levels/behaviours
- Provision of Information for further assistance: Pamphlets/contact numbers for drug and alcohol related community services

CASE THREE

Short case number: 4_12_03

Category: Clinical skills_Pharmacology / Mental health & Human Behaviour

Discipline: Emergency Medicine

Setting: Emergency Department_rural

Topic: Overdose_Principles

CASE



A 25yo female presents to ED by ambulance. She was found at home by her friend with several empty tablet blister-packs nearby. She had recently ended a long-term relationship.

On arrival at ED she has a decreased level of consciousness, a BP of 70/40, heart rate 140/min with multiple ectopic beats. There are no signs of trauma.

The blister-packs contained a mixture of antidepressant and anxiolytic drugs including Citalopram, Doxepin, Lorazepam, and Moclobemide.

QUESTIONS

1. Outline the key components of the assessment and initial management of this patient.
2. What is the role of activated charcoal in the management of overdose?
3. Often overdoses are caused by the so called 'street' and 'party' drugs - Describe the typical presentations and management principles of overdose of common "street" and "party" drugs.
4. Under what circumstances would extracorporeal drug removal be appropriate, eg haemodialysis?
5. Outline the toxidromes that occur with overdose of antidepressant drugs such as in this case with OD on Tricyclic, Monoamine Oxidase inhibitor, and Serotonin Reuptake inhibitors?

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

Outline the key components of the assessment and initial management of this patient.

ABC – resuscitation. In this case, high-flow oxygen initially with consideration to intubation to prevent aspiration and to ensure adequate ventilation. Venous access and an initial fluid bolus of either 1000ml Hartmann's solution or 500ml colloid solution. If VEB's progress to cardiac arrest, manage as per ARC guidelines. Bloods for UEC, LFT, ABG, Urine Drug screen. Anion Gap and Osmolar gap. CXR (aspiration). Specific drug levels if appropriate (eg Lithium; paracetamol; digoxin; salicylates; Fe)

Rule out other causes of unconsciousness.

The majority of poisonings present with one or more of : haemodynamic disturbance; decreased LOC; temperature disturbance; GIT disturbance.

Administration of specific antidote if applicable. Examples : Naloxone (opioids); n-Acetyl Cysteine (Paracetamol); Ethanol (methanol); Hyperbaric Oxygen (Carbon monoxide); Fab-Digoxin antibodies; Atropine and

What is the role of activated charcoal in the management of overdose?

Prevent further drug absorption (skin, GIT). Skin absorption especially important with organophosphates (remove and dispose of all clothing; precautions for staff as well). GIT means induce vomiting (no longer recommended; never with volatile hydrocarbons because of risk of aspiration and pneumonitis; never if patient has depressed LOC) and oral Activated Charcoal . Give either via NG tube or to drink. Requires intubation if decreased LOC. Remember that in the majority of poisonings, aspiration is more likely to cause death than the poisoning. Produced by superheating charcoal. Very fine particles hence huge surface area. Adsorbs drug. Often given multiple doses q6h for drugs with enterohepatic circulation eg Theophylline, Carbamazepine; Digoxin; and for sustained-release preparations.. Often with sorbitol as a cathartic.

Describe the typical presentations and management principles of overdose of common "street" and "party" drugs.

Common street and Party drugs:

- Heroin. Unconscious, pinpoint pupils, respiratory depression. May have cardio-resp arrest. Manage with ABC/Naloxone 400mcg ivi, 2mg imi as risk of re-narcotisation with iv dose alone. Beware HIV/HepC.
- Cocaine and Amphetamines/Ice. Hypertension, arrhythmias. May develop cerebral haemorrhage and AMI. Treat hypertension with vasodilators, avoid B-Blockers due to risk of unopposed α -activity.

- c) GHB. Unconscious, haemodynamically reasonably stable, bradycardia, small pupils, hypothermia. Management entirely supportive.
- d) Ecstasy. Typical presentation with hyponatraemia and seizure. ABC, correct sodium, terminate seizure with benzodiazepines, supportive care. Hyponatraemia due to loss of large amounts of Na-rich fluid (sweat in particular) and replacement with water alone. Rapid correction is fine for these patients ie no need to correct slowly as the problem almost certainly developed very rapidly..
- e) Benzodiazepines. Decreased LOC. Often taken with alcohol and other drugs. Respiratory depression. Flumazenil can be given but risk of acute withdrawal with seizures. Manage with supportive care.

Under what circumstances would extracorporeal drug removal be appropriate, eg haemodialysis?

Dialysis. Only for drugs/toxins that are : largely non-protein-bound; present in large quantities; have small volume of distribution (ie are not stored in fat etc) and have significant risk of major toxicity even with good supportive care. Examples include : Ethylene Glycol (anti-freeze); Methanol: **For the vast majority of poisonings, good supportive care is all that is required.** Attention to maintenance of normal haemodynamics, respiratory support, DVT prevention, pressure area prevention, prevention of aspiration, etc

Outline the toxidromes that occur with overdose of antidepressant drugs such as in this case with OD on Tricyclic, Monoamine Oxidase inhibitor, and Serotonin Reuptake inhibitors?

Tricyclics. In particular, malignant ventricular arrhythmias, hypotension, anticholinergic effects (CNS and peripheral). MAOI's : Moclobemide is a reversible MAOI but the clinical picture would be similar to traditional irreversible MAOI poisoning, with hypotension (they were originally used as antihypertensives!) and catastrophic hypertension from administration of indirectly-acting amines such as ephedrine and metaraminol, also Pethidine. SSRI's in overdose, and particularly if there is also tricyclic antidepressant intoxication, can present with the "Serotonin Syndrome" with hyperthermia, coma, rigidity, leading to death.

CASE FOUR

Short case number: 4_12_04

Category: Metabolism

Discipline: Emergency Medicine

Setting: Emergency Department

Topic: Disorders of Temperature Hyperthermia

CASE 1



A 32 year old male is brought to the emergency department following running in a marathon on a 42°C day.

He is confused with a GCS of 13 and has a pulse rate of 114 and BP of 98/60 and temperature of 43°C.

He has no allergies, and no significant medical or surgical history. He is not on any medications.

QUESTIONS

1. Outline the possible diagnoses.
2. What are the important components of assessment and management of this patient?
3. Outline the necessary investigations in this patient
4. Outline your immediate management of this patient

ANSWERS

Outline the possible diagnoses.

The main concern in this case is a heat related illness and especially heat stroke. The classic triad of neurological abnormality, hyperthermia(>41C) and lack of sweating may not always be present. Mortality from this condition is very high at around 70+% and so rapid treatment is necessary.

What are the important components of assessment and management of this patient?

Outline the necessary investigations in this patient

Outline your immediate management of this patient

The patient should be in a resuscitation cubicle.

There is a possibility of worsening conscious state and even seizure. Intubation and airway management should occur as needed.

Patients are usually dehydrated with an up to 2-3 litre deficits.

Give normal saline and monitor status.

If vasopressors are needed, care should be taken with the use adrenaline, as this can increase peripheral vasoconstriction resulting in reduced heat lost from the body.

The patient should then be cooled.

There are several types of cooling methods, from least to very invasive.

The most popular techniques are: ice packs to axillae and groin and mist water with fan cooling../ More invasive methods of gastric and peritoneal lavage are available but may not be appropriate. Beware to cease cooling at about 38C as can overshoot.

Maintain good urine output at 1-2mL/kg/hour in patient with renal damage or rhabdomyolysis.

Because of this potential muscular destruction and rhabdomyolysis, care should be taken with the use suxamethonium when intubating, unless a potassium is known, as the patient may be hyperkalaemic. Instead non-depolarising muscle relaxants should be used.

Blood test should include FBC, EUC and LFT looking at renal function and liver function as AST>1000IU in first 24 hours in association with continued hyperthermia and hypotension are poor prognostic indicators.